

Airbnb Price Prediction in NYC

Executive Summary

Project: Airbnb Market Analysis – New York City (2019)

Dataset: AB_NYC_2019.csv

Objective: To analyze the NYC Airbnb market dynamics, identify key factors influencing listing prices, and build a predictive pricing model.

Background

Airbnb has revolutionized the hospitality industry by facilitating peer-to-peer short-term rental transactions, reshaping how travelers experience cities worldwide. As one of the largest and most competitive markets globally, New York City stands out for its diverse neighborhoods and dynamic rental landscape.

Understanding the key factors influencing Airbnb pricing in NYC is essential for multiple stakeholders—hosts aiming to optimize their revenue, investors seeking lucrative opportunities, and city planners monitoring housing trends and tourism impacts.

Problem Statement

The short-term rental market in New York City is highly competitive and influenced by a multitude of factors including location, property type, and guest reviews. However, price setting remains a challenge for hosts due to the complex interplay of these variables and market dynamics. Without a thorough understanding of what drives pricing, hosts risk underpricing or overpricing their listings, leading to lost revenue or lower occupancy rates. This study aims to identify and quantify the key determinants of Airbnb listing prices in NYC, providing actionable insights to help stakeholders set optimal prices that balance competitiveness and profitability.

Size: ~48,895 listings

Features: 'id', 'name', 'host_id', 'host_name', 'neighbourhood_group', 'neighbourhood', 'latitude', 'longitude', 'room_type', 'price', 'minimum_nights', 'number_of_reviews', 'last_review', 'reviews_per_month', 'calculated_host_listings_count', 'availability_365'

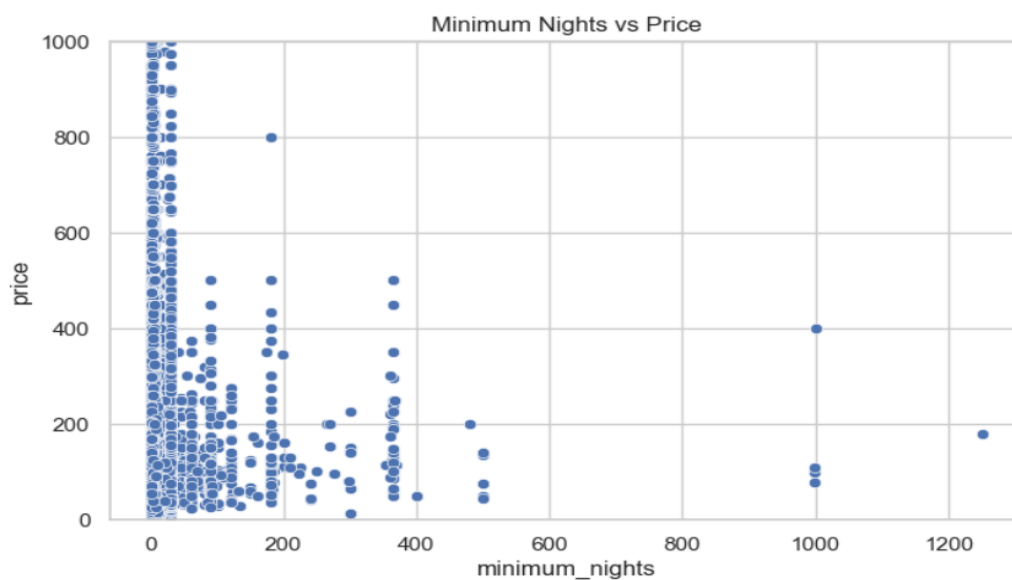
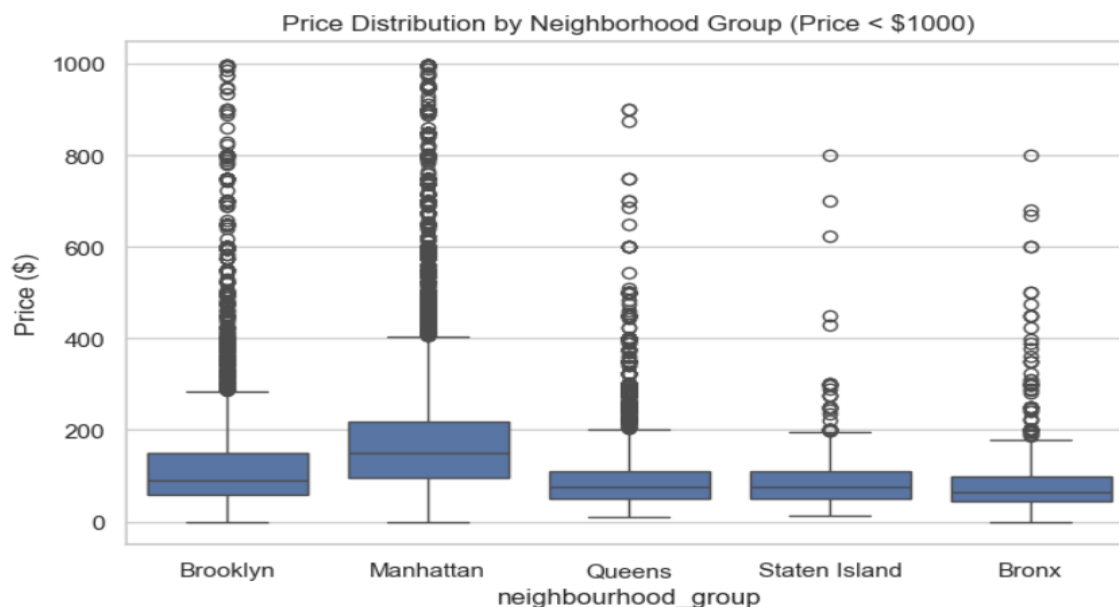
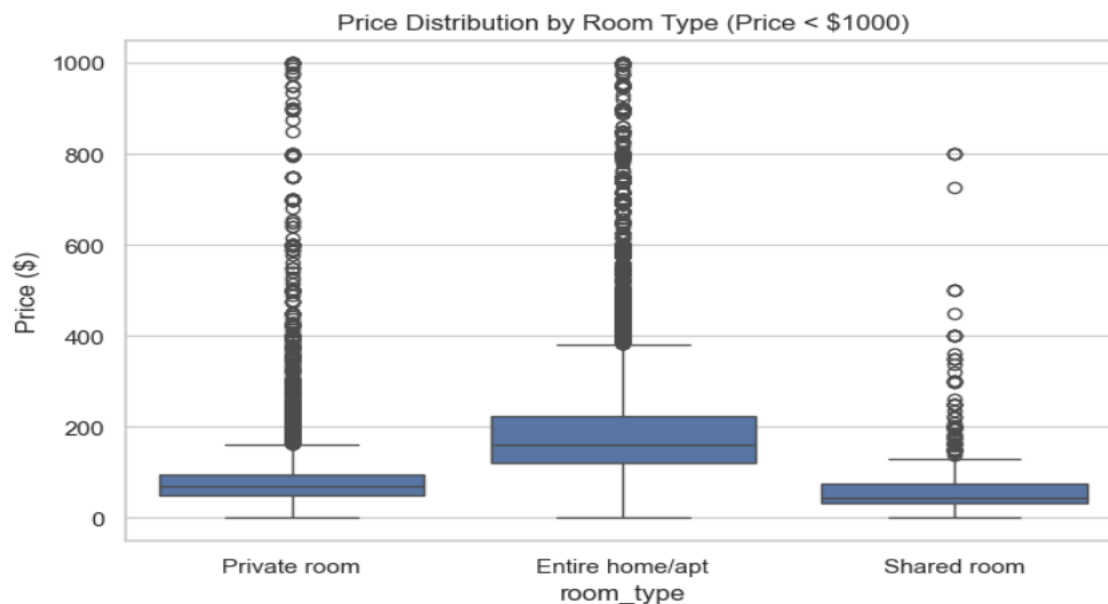
Geography: Five boroughs of NYC, including Manhattan, Brooklyn, Queens, Bronx, and Staten Island

Key Findings

- Price Distribution:** Prices are right-skewed with most listings under \$200. Outliers were removed (> \$1000) for stable modelling. This showcased higher percentage of customers were preferring cheaper stays.



- **Price Drivers:** Room type, neighbourhood group (borough), minimum nights, and review activity are significant predictors of price. Entire homes command the highest prices, followed by private rooms.



- **Geospatial Insights:** Manhattan and Brooklyn neighbourhoods show higher average prices compared to other boroughs. Location clusters derived from latitude and longitude improve pricing prediction accuracy.

Average Price by Neighborhood Group (Borough):

```
neighbourhood_group
Manhattan      196.875814
Brooklyn       124.383207
Staten Island  114.812332
Queens         99.517649
Bronx          87.496792
Name: price, dtype: float64
```



Data Cleaning Approach

To ensure reliable and accurate analysis, the Airbnb NYC dataset underwent a thorough data cleaning process. First, listings with extreme price values above \$1,000 were excluded to remove outliers that could distort modeling. Missing values, particularly in critical fields such as reviews_per_month, were handled by removing records with null entries to maintain data integrity. Categorical variables such as room_type and neighbourhood_group were encoded appropriately to facilitate quantitative analysis. This rigorous cleaning process ensured a high-quality dataset conducive to robust price prediction and insightful exploratory analysis.

Choice of Model: Why Random Forest?

1. **Handling of Non-linear Relationships:** Airbnb pricing is influenced by complex, non-linear interactions between features such as location, room type, and availability. Random Forest naturally captures these patterns without requiring extensive feature engineering.
2. **Resistance to Outliers and Noise:** The model is robust to outliers and noisy data, which are common in real-world datasets like Airbnb listings.
3. **Feature Importance Insights:** Random Forest provides measures of feature importance, offering valuable business insights into which factors most strongly drive pricing.
4. **Ease of Use and Scalability:** It requires minimal parameter tuning and performs efficiently on moderate-sized datasets like this Airbnb dataset.

Business Takeaways

- **Price Optimization:** Hosts can leverage insights on key price drivers such as room type and neighborhood to competitively price their listings and maximize revenue. Entire homes and listings in Manhattan or Brooklyn generally command premium rates.
- **Targeted Marketing:** Identifying neighborhoods with higher average prices allows property managers and investors to focus marketing efforts on high-demand areas, enhancing booking rates and profitability.
- **Inventory Management:** Listings with low minimum nights and active review profiles tend to perform better. Hosts should consider flexible stay requirements and encouraging guest feedback to improve visibility and attract more bookings.
- **Investment Decisions:** Real estate investors can use location clusters and pricing patterns to make informed decisions on property acquisitions for short-term rentals, balancing cost and return.
- **Policy and Planning:** City planners and regulators may utilize pricing trends and neighborhood data to better understand the short-term rental market's impact on housing affordability and tourism.

Recommendations from domain perspective

- **Enhance Listing Quality:** Hosts should invest in high-quality photos, detailed descriptions, and amenities to improve guest experience and justify higher pricing.
- **Dynamic Pricing Tools:** Use automated pricing tools that adjust nightly rates based on seasonality, local events, competitor prices, and demand fluctuations to maximize occupancy and revenue.
- **Encourage Reviews:** Positive guest reviews boost listing credibility and search ranking. Hosts should engage with guests, respond to feedback, and encourage reviews to maintain a strong reputation.
- **Flexible Policies:** Offering flexible cancellation and minimum night policies can attract a wider range of travelers and increase booking frequency.
- **Leverage Data Analytics:** Property managers and investors should routinely analyze market trends, competitor pricing, and guest preferences to adapt their strategies proactively.
- **Compliance & Community Engagement:** Ensure compliance with local regulations to avoid penalties. Engage with local communities to promote sustainable tourism and mitigate potential neighborhood concerns.
- **Diversify Listings:** Consider diversifying property types and locations within a portfolio to reduce risk and capture different market segments.
- **Sustainability Initiatives:** Promote eco-friendly practices and certifications to appeal to environmentally conscious travelers and differentiate listings.

My Learnings

Working on this Airbnb price prediction project has been a truly eye-opening experience for me. Right from the beginning, I realized how important data cleaning and feature engineering are in building a robust model. I spent considerable time exploring the dataset, handling missing values, and removing outliers to make sure the model had clean, reliable data to learn from. Creating new features like location clusters from latitude and longitude helped me understand how domain knowledge can enhance predictive power. As I dug deeper into the data, I came face-to-face with the complexities of real-world data — like the skewed price distribution and the challenges of dealing with categorical variables such as room types and neighborhoods. This made me appreciate that data science isn't just about algorithms; it's about truly understanding the data and how it reflects the problem at hand.

What really stood out for me was how technical work can directly translate into business value. Through this project, I could see how data-driven insights about pricing, neighborhoods, and listing features can empower hosts to optimize their pricing and help investors make smarter decisions. It reinforced my belief that bridging the gap between data science and business is essential for impactful solutions.

Finally, building and deploying the interactive Streamlit app was a rewarding experience. It made me think about the end-user and the importance of creating tools that are accessible and easy to use. This helped me realize that no matter how good a model is, its real value lies in how effectively people can interact with and apply it.

Overall, this project has strengthened not only my technical skills but also my ability to think critically about data, business applications, and user experience. It has been a great learning journey that I'm excited to build upon in future projects.

